

SPX Student Case Data Dictionary

Synthetic logistics data pack for university hackathon analysis

Domain Primer

Term	Definition
SPX	A fictionalized parcel delivery network used in this student case.
Hub	A local facility where parcels are sorted and dispatched for last-mile delivery.
Rider	A delivery worker assigned to bring parcels from a hub to customers.
Backlog	Parcels that remain waiting after the normal sort or dispatch cutoff.
Retry / redelivery	A parcel that required more than one delivery attempt.
Failed delivery	A parcel not successfully delivered on the recorded attempt window.
Late delivery	A parcel whose actual delivery days exceeded the promised days.

Files Included

parcel_history.csv has one row per parcel outcome. hub_daily_operations.csv has one row per hub per day. cost_assumptions.xlsx contains rough cost, value, and solution-menu assumptions for business-case sizing.

Synthetic Design Notes

- HUB_C_SOUTH is intentionally more exposed to rider load and backlog stress on peak days.
- HUB_E_WEST has a stronger address-quality and customer-availability pattern.
- HUB_F_RIVER is more sensitive to weather, traffic, and vehicle availability disruptions.
- The data includes noise and overlapping drivers, so students should compare evidence rather than hunt for one perfect answer.

Column Dictionary

Column	Definition	Type	Allowed values / units
parcel_id	Unique synthetic parcel identifier.	Text	P0000001 style
hub_id	Fictional hub responsible for last-mile delivery.	Text	HUB_A_NORTH, HUB_B_CENTRAL, HUB_C_SOUTH, HUB_D_EAST, HUB_E_WEST, HUB_F_RIVER
date	Calendar date for parcel outcome or hub operating day.	Date	yyyy-mm-dd
origin_zone	Broad pickup/origin zone.	Category	Metro, North, South, East, West, River
destination_zone	Broad destination zone.	Category	Metro, North, South, East, West, River, Remote
service_type	Delivery promise tier.	Category	standard, express, economy
seller_type	Simplified seller segment.	Category	marketplace, direct_seller, small_business, enterprise
parcel_size_band	Size category used for handling complexity.	Category	XS, S, M, L, XL
parcel_weight_band	Weight category used for handling complexity.	Category	0-1kg, 1-3kg, 3-5kg, 5-10kg, 10kg+
promised_delivery_days	Days promised to customer.	Number	1, 2, 3
actual_delivery_days	Actual days to final outcome in this synthetic case.	Number	Whole days
delivery_outcome	Final parcel status for analysis.	Category	delivered_on_time, delivered_late, failed
failure_reason	Primary reason if delivery_outcome is failed.	Category	customer not available, bad address, rider capacity, routing delay, hub backlog, weather/disruption
attempt_count	Delivery attempts recorded for the parcel.	Number	1 to 4
is_redelivery	Whether more than one attempt was needed.	Boolean	TRUE/FALSE
was_late	Whether actual delivery exceeded promise.	Boolean	TRUE/FALSE
customer_contact_made	Whether the operation contacted the customer.	Boolean	TRUE/FALSE
priority_flag	Whether the parcel received priority handling.	Boolean	TRUE/FALSE
peak_day_flag	Whether the date had elevated volume pressure.	Boolean	TRUE/FALSE
region	Broad operating region for the hub.	Category	Metro/Province/River corridor labels
hub_type	Hub operating environment.	Category	urban, suburban, mixed
daily_parcel_volume	Parcels represented for the hub that day.	Number	Parcel count
daily_successful_deliveries	Delivered parcels that did not fail.	Number	Parcel count
daily_failed_deliveries	Failed delivery parcels.	Number	Parcel count
daily_late_deliveries	Parcels flagged late, including some failed attempts.	Number	Parcel count
active_riders	Riders available for delivery that day.	Number	Headcount
available_sort_staff	Sort staff available at the hub that day.	Number	Headcount
overtime_hours	Estimated overtime used that day.	Number	Hours

Column	Definition	Type	Allowed values / units
avg_parcels_per_rider	Daily workload per active rider.	Number	Parcels/rider
avg_sorting_backlog	Average parcels waiting after sorting cutoff.	Number	Parcel count
avg_dispatch_delay_hours	Average delay from ready-to-dispatch to dispatch.	Number	Hours
vehicle_availability_rate	Share of planned vehicles available.	Percent	0.00 to 1.00
traffic_disruption_flag	Whether traffic disruption affected the hub.	Boolean	TRUE/FALSE
weather_disruption_flag	Whether weather or local disruption affected the hub.	Boolean	TRUE/FALSE
temporary_staff_used	Whether temporary staff were used.	Boolean	TRUE/FALSE
customer_complaints_count	Customer complaints attributed to the hub/date.	Number	Count

Example Business Questions

- Which hub or operating condition seems most linked to poor delivery outcomes?
- Are delays driven more by volume, staffing, disruptions, or specific parcel types?
- Which failure reasons are most controllable by SPX in the next 30 to 60 days?
- Which first solution should SPX test, and in which hub should it be piloted?
- How would students measure whether the pilot worked?

Suggested Metrics

Metric	Simple calculation
Failure rate	$\text{daily_failed_deliveries} / \text{daily_parcel_volume}$
Late rate	$\text{daily_late_deliveries} / \text{daily_parcel_volume}$
Workload	$\text{avg_parcels_per_rider}$
Backlog pressure	$\text{avg_sorting_backlog} / \text{daily_parcel_volume}$
Complaint rate	$\text{customer_complaints_count} / \text{daily_parcel_volume}$
Retry rate	$\text{Count of is_redelivery TRUE} / \text{parcel count}$

Important Caveat

This is synthetic case data, designed for learning. It should not be treated as real SPX operating data, causal proof, or finance-grade planning input.